Movie Recommendations

Creators: Cameron, Ridge, Tim

Movie Data

- MovieId
- Title
- Genres

Ratings Data

- userId
- MovieId
- Rating (target variable)
- Timestamp

Only 100,000

Modeling Challenge

- We want to be able to predict the rating a user gives an arbitrary movie.
- How can we quantify the 'link' between a user and a movie?

- Solution: Two Towers!
- Dimensionalize the users and movies in the same space

Two Towers Method

User ID "Tower"

Ratings

Movie ID "Tower"

Orthanc

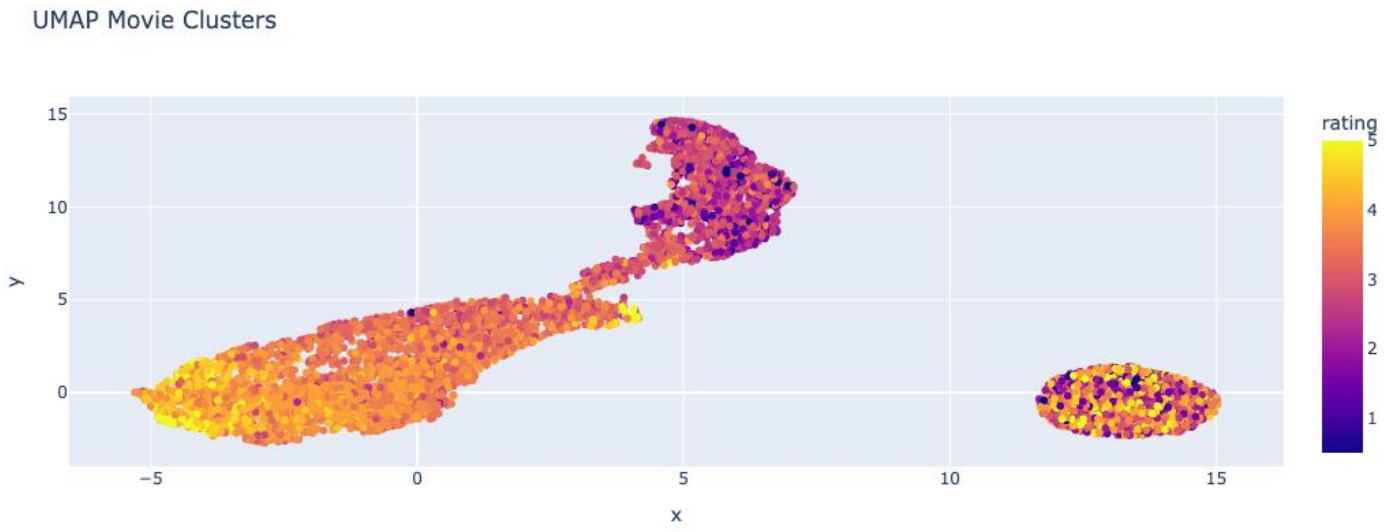
Neural Network

32D mapping of both towers

Minas Morgul

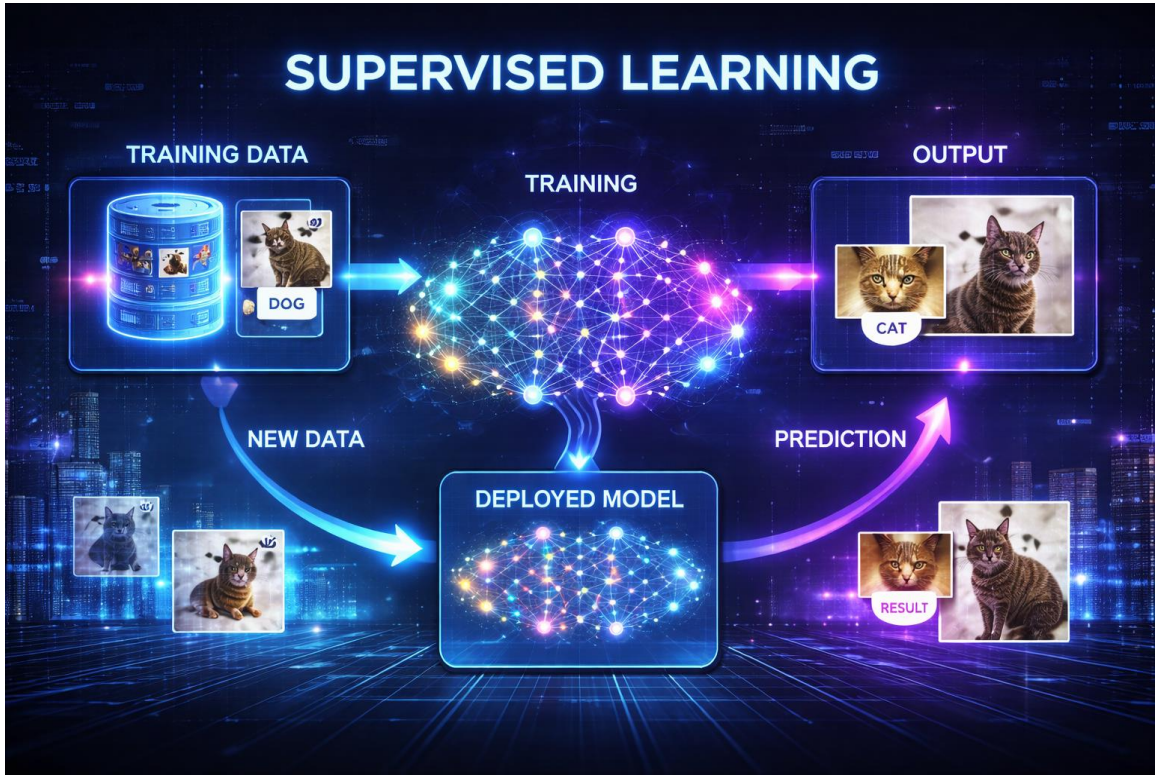
Two Towers Usage

- Use UMAP to visualize how 'similar' movies are
- Using our embedded data, we can calculate an affinity score between users and movies (cosine distance).



Supervised Model Performance

- NN Model
  - F1 Score: .767 (higher predictive power)
  - Better at continuous prediction between relationships
- KNN Model
  - F1 Score: .679 (moderately high predictive power)
  - Better at capturing the nonlinear relationships



Next Steps & Improvements

Case-by-case recommendation

Allow streamlit users to input their own data

The most exciting/fun potential feature

Difficult to add/create to the repo

Model Enhancements

Use Genre importance more

Improve recommender performance

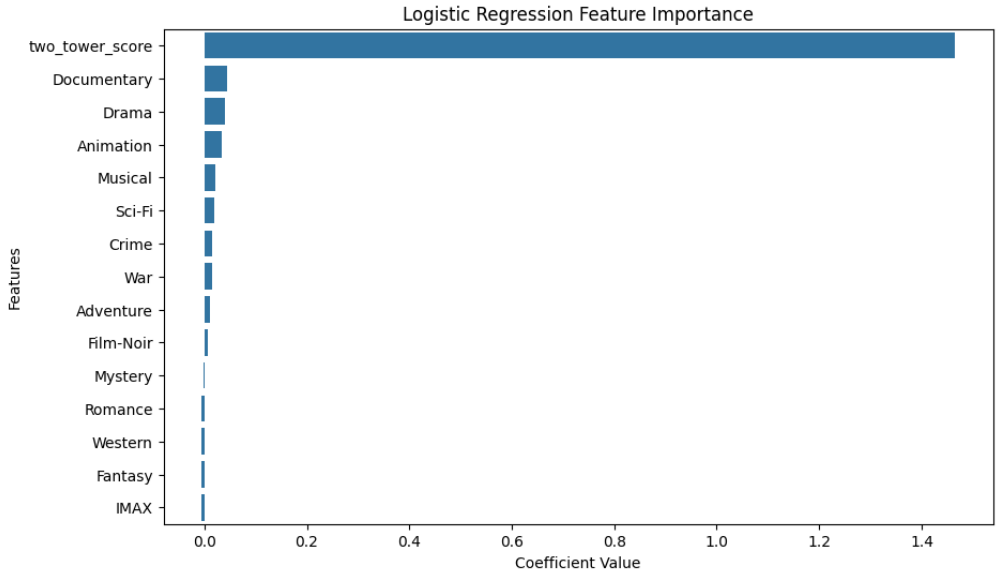
Explore ranking models

Improve our feature set

Feature Engineering

User activity patterns

More metadata (tags, other data)



Demonstration

- https://stat486finalproj.streamlit.app/

Thank you!